

DTSA 5509

Introduction to Machine Learning: Supervised Learning

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Data Scientist (60.8 / 100)	49.6 / 100
SPECIALIZATION Machine Learning: Theory and Hands-on Practice with Python		

50

CURRENT AVG SCORE
across 6 job roles

81

POTENTIAL SCORE
with recommended additions

+41

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces various supervised ML algorithms and prediction tasks applied to different data. Specific topics include linear and logistic regression, KNN, Decision trees, ensemble methods such as Random Forest and Boosting, and kernel methods such as SVM. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	60.8	98.8	+38.0	HIGH
ML Engineer	60.3	79.5	+19.1	MEDIUM
Machine Learning Engineer	54.2	85.8	+31.5	HIGH
AI Engineer	44.7	85.8	+41.1	HIGH
Applied Scientist	43.0	71.9	+28.9	MEDIUM
MLOps Engineer	34.5	62.8	+28.3	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Data Scientist

60.8 → 98.8 (+38.0 pts)

product mitigation behaviors build

ML Engineer

60.3 → 79.5 (+19.1 pts)

learning machine systems models model build product new

Machine Learning Engineer

54.2 → 85.8 (+31.5 pts)

interfaces predictive learning tools design development science models

AI Engineer

44.7 → 85.8 (+41.1 pts)

models learning build technical model systems machine design

Applied Scientist

43.0 → 71.9 (+28.9 pts)

models technical learning systems design development regulatory science

MLOps Engineer

34.5 → 62.8 (+28.3 pts)

model text systems learning machine tools models development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.